Algorithm 2 A* search on the grid, $h(s) = \|s - s_{\text{goal}}\|$

```
function AStar(s_start, s_goal, w)
    g(s_start) ← 0;  OPEN ← {s_start};  CLOSED ← ∅
    while OPEN ≠ ∅ do
        s ← arg min_{s∈OPEN}  g(s) + w · h(s)
        if s = s_goal then
            return ReconstructPath(parent, s_goal)
        end if
        OPEN ← OPEN \ {s};  CLOSED ← CLOSED ∪ {s}
        for all s' ∈ Neighbors(s) do
            if s' ∈ CLOSED ∨ ¬CollisionFree(s, s') then
                continue
            end if
            g' ← g(s) + Cost(s, s')
            if s' ∉ OPEN ∨ g' < g(s') then
                g(s') ← g';  parent(s') ← s
                OPEN ← OPEN ∪ {s'}
            end if
        end for
    end while
    return failure
end function
```